Class XII Session 2025-26
Subject - Chemistry
Sample Question Paper - 4

Time Allowed: 30 Minutes.

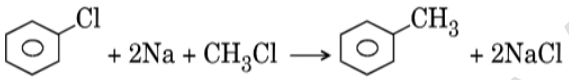
General Instructions:

Read the following instructions carefully.

1. There are 13 questions in this question paper with internal choice.
2. SECTION A consists of 13 multiple-choice questions carrying 1 mark each. **[1]**
3. **All questions are compulsory.**
4. **Use of log tables and calculators is not allowed.**

Section A

1. The reaction given below:

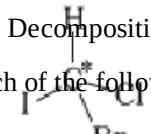


a) Wurtz reaction
b) Gattermann reaction
c) Wurtz - Fittig reaction
d) Fittig reaction

2. Which among the following is an example of pseudo first order reaction? **[1]**

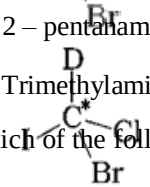
a) Decomposition of hydrogen peroxide
b) Dehydration of oxalic acid
c) Decomposition of nitrogen pentoxide
d) Acid catalysed hydrolysis of ethyl acetate

3. Which of the following amines can be resolved into enantiomers? **[1]**



a) 2 – pentanamine
b) 3 – pentanamine
c) Trimethylamine
d) 4 – (dimethylamino) pyridine

4. In which of the following molecules carbon atom marked with an asterisk (*) is asymmetric? **[1]**



a.

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b.

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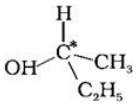
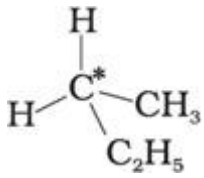
c. 

d. 

a) (a), (b), (c), (d)

b) (b), (c), (d)

c) (a), (c), (d)

d) (a), (b), (c)

5. Alkenes react with water in the presence of acid as catalyst to form alcohols. **[1]**

a) Deprotonation to form alcohol

b) nucleophilic attack of water on carbocation

c) Protonation of alkene and carbocation

d) All of these

6. Match the items of column I with appropriate entries of column II. **[1]**

Column I	Column II
(a) K_b	(i) Elevation in boiling point
(b) K_f	(ii) Van't Hoff factor
(c) i	(iii) Cryoscopic constant
(d) $\triangle T_b$	(iv) Ebullioscopic constant

a) (a) - (ii), (b) - (i), (c) - (iii), (d) - (iv)

b) (a) - (i), (b) - (ii), (c) - (iii), (d) - (iv)

c) (a) - (iii), (b) - (iv), (c) - (i), (d) - (ii)

d) (a) - (iv), (b) - (iii), (c) - (ii), (d) - (i)

7. How many alcohols with molecular formula $C_4H_{10}O$ are chiral in nature? **[1]**

a) 1

b) 3

c) 2

d) 4

8. Give the products of the reaction **[1]**

$$PhC \equiv CMe \xrightarrow{H_3O^+, Hg^{2+}} ?$$

a) PhCOCOMe

b) $PhCH_2COCH_3$

c) $PhCOCH_2CH_3$

d) $PhCH_2CH_2CHO$

9. On addition of small amount of $KMnO_4$ to concentrated H_2SO_4, a green oily compound is obtained which is highly explosive in nature. Identify the compound from the following. **[1]**

a) MnO_2

b) Mn_2O_7

c) Mn_2O_3

d) $MnSO_4$

10. The major component of starch is: **[1]**

a) water

b) glucose

b) amylopectin

d) amylose

11. The molecular formula of ethyl acetate is:

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a) $C_5H_8O_2$

b) $C_5H_{10}O_2$

c) $C_4H_8O_2$

d) C_4H_8O

12. Which among the following is an example of photochemistry used in our daily life? **[1]**

a) In inversion of cane sugar

b) In photography

c) In decomposition of hydrogen peroxide

d) All of these

13. **Assertion (A):** 2-Butanol on heating with H_2SO_4 gives 1-butene and 2-butene. **[1]**

Reason (R): Dehydration of 2-butanol follows Saytzeff's rule.

a) Both A and R are true and R is the correct explanation of A.

b) Both A and R are true but R is not the correct explanation of A.

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c) A is true but R is false.

d) A is false but R is true.

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